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Current status of public education on endocrine and metabolic diseases carried out by academic journals' WeChat official accounts: a retrospective study

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Abstract

Background To investigate the current status of public education on endocrine and metabolic diseases through WeChat official accounts (WOAs) of academic medical journals in China and to identify the factors that influence user engagement.

Methods A retrospective study was conducted to assess the effectiveness of WOAs from both general medical journals (GMJs) and endocrinology medical journals (EMJs) listed in the Chinese Science Citation Database. After characterising the main features of the articles, we used generalized linear models to identify factors associated with the number of readings and likes of the articles. .

Results The study identified 16 WOAs of medical journals, consisting of 12 GMJs and 4 EMJs. From 2021 to 2023, these WOAs published a total of 1,033 articles on endocrine and metabolic diseases. The most popular topics were diabetes (680/1033, 65.8%), osteoporosis and metabolic bone disease (168/1033, 16.2%), and thyroid disease (62/1033, 6.0%). Multivariable regression analysis revealed that in GMJs, longer articles and those categorized as guideline/interpretation and expert consensus positively predicted both reading and liking counts. In EMJs, longer articles and guideline/interpretation and expert consensus categories were associated with increased reading counts, while guideline/interpretation and expert consensus articles also positively predicted liking counts.

Conclusion Although articles on endocrine and metabolic diseases achieve relatively high engagement on WOAs of academic medical journals in China, the overall reads and like counts remain low. Further optimization of promotion strategies, especially for incorporating multimedia content, is needed to significantly enhance user engagement.

Keywords WeChat, Official account, Medical academic journals, Endocrine and metabolic diseases

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Introduction

The past decade has witnessed a fundamental transformation in academic dissemination, driven by the rise of new media platforms. As a traditional media outlet, medical journals carry the vital responsibility to propagate medical knowledge and research advancements. Developments in information technology and social media have continuously reshaped journals' publishing and dissemination. Knowledge dissemination has evolved from text-based formats to dynamic multimodal approaches incorporating text, audio, and video. The 2024 Research Report on the Development of Online Audiovisual in China reveals that online audio-visual users in the country exceeded 1.074 billion by December 2023, representing 98.3% of internet users. This remarkable growth underscores how medical journal dissemination has entered the new media era, rapidly expanding digital content significantly accelerating medical education [1].

English-language medical journals have proactively embraced new media for disseminate knowledge. The medical community widely utilized social media platforms, including Twitter and Facebook, to share insights and educate patients [2, 3]. Leading international journals, such the Lancet and the New England Journal of Medicine, maintain official accounts on these platforms, actively promiting medical knowledge, expanding their readership, and enhancing content visibility [4, 5]. This approach significantly amplifies their global impact within the medical community.

In recent years, the WOA has emerged as a highly influential new media platform in China, distinguished by the convenience, interactivity, and personalized content delivery [6]. This versatile platform enables targeted dissemination of diverse media formats, facilitating one-to-many communication and cementing its role as a vital tool in health education [7]. Hospitals across China have embraced WOAs showcase medical service and engage in health education activities [8, 9]. Similarly, disease control centers utilizes these accounts to disseminate health knowledge, accoss topics such as nutrition, environmental sanitation, and the prevention and management of infectious diseases [10, 11]. Notably, during the Covid-19 pandemic, disese control centers' WOAs played a pivotal role in public health education [12, 13]. Research indicates that health education via WeChat platforms significantly enhances health literacy among patients and their families [14, 15]. Recognizing new media's poential, academic journals have begun exploring WOAs for health education. An increasing number of Chinese journals now maintain WOAs, presenting educational content through diverse formats, such as images, texts, and videos. Furthermore, the WOA platform is distinguished by robust interactive capabilities and efficient dissemination mechanisms, enabling rapid spread of health education information to a broader audiences.

Endocrine and metabolic diseases constitute a substantial global health burden, characterized by high prevalence, chronic progression, and severe complications that significantly impair patients' physical and mental well-being. For example, diabetes prevalance in China increased from 10.9% in 2013 to 12.4% in 2018 [16], with patient number projected to reach 174 million by 2045. Among Chinese adults aged 40 years and above, osteoporosis prevalence is 5% in males and 20.6% in females [17], rising to 6.46% and 29.13% respectively in those over 50 [18]. Additionally, thyroid nodules, with an overall prevalence of 36.9%, have emerged as a notable public health concern [19]. Given these high incidence rates and the chronic nature of these disease, strengthening public health education is imperative.

In this retrospective study, we evaluated the status and user engagement of health education on endocrine and metabolic diseases conducted by academic journals through the WOA platform.

Methods

Study design and data source

We searched the WOAs of Chinese medical journals through the WeChat APP and further collected posted articles on endocrine and metabolic diseases from January 1, 2021, to December 31, 2023. The selection criteria of Chinese medical journals were as follows:

1) Journals were chosen from the Chinese Science Citation Database (CSCD) list, encompassing both GMJs and EMJs;
2) The journal must have an officially certified WOA.

The exclusion criteria were:

1) Medical journals from other disciplines, such as surgery
2) Journals focused on traditional Chinese medicine;
3) Journals of Stomatology;
4) Pharmacy journals;
5) Public health journals;
6) Basic medical research journals;
7) Nursing journals
8) Journals whose WOAs were not officially certified;
9) WOAs that only posted articles in collections without individual article posting.

The flowchart of this study is shown in Fig. 1.

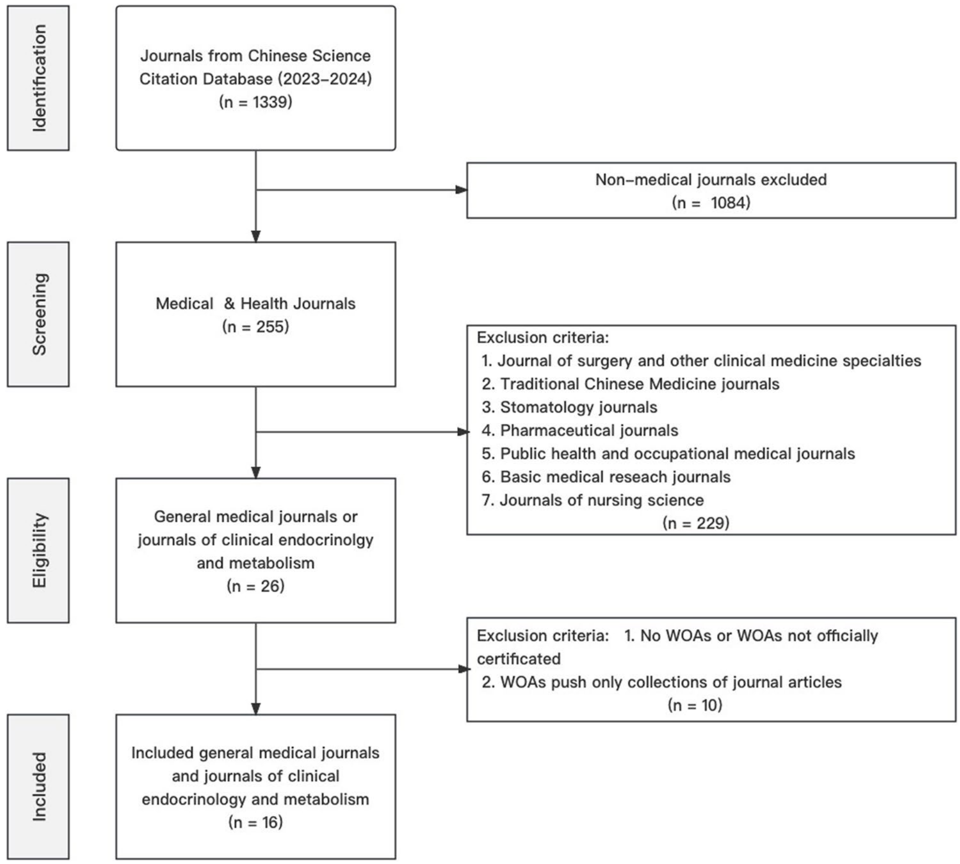

Fig. 1 The flowchart of the study

Main parameters and article features frame

Initially, we conducted a thorough literature review and established the key parameters for article evaluation [20]. The recorded parameters included:

1) Article content: Eight categories were identified, covering adrenal disease and hypertension, diabetes, gonadal disease, neuroendocrine tumors, obesity, osteoporosis and metabolic bone disease, pituitary disease, and thyroid disease.
2) Article order: The sequence in which articles were posted.
3) Article format: Categorized in to three distinct formats, namely "text only", "text + pictures", and "text + pictures + videos".
4) Length of the article: Divided into three groups based on word count, length < 1k words, length 1k-5k words, length > 5k words.
5) Article category: Including reviews, basic research articles, clinical research articles, case reports, Mendelian randomization analyses, meta-analysis and systemic reviews, guidelines and their interpretation, expert Consensus documents, and others.

The framework and main features are detailed in Table 1. User engagement was assessed using the metrics of reading and liking. The data collection process was conducted independently by two researchers, followed by a cross-checking procedure to ensure accuracy and consistency.

Statistical analysis

Continuous variables are presented as medians with interquartile ranges (IQRs). Normality was assessed using the Shapiro-Wilk test. Group comparisons were performed using one-way analysis of variance for variables that met the assumption of normality, whereas the Kruskal–Wallis test was applied when the normality assumption was not satisfied. Categorical variables are summarized as frequencies and percentages. To identify factors associated with article readings and likes, generalized linear models (GLMs) were fitted separately for each outcome. Given the count nature and potential overdispersion of the data, quasi-Poisson regression with a log link function was employed. Covariates included article format, length, category, content, and posting order. Analyses were stratified by journal type (general medical journals and endocrinology and metabolism journals). Regression coefficients with 95% confidence intervals (CIs) and P values are reported. All analyses

Table 1 The main features of articles with definitions

Item	Definition
Article content	
1) Adrenal disease and hypertension	Topic is related to adrenal disease and hypertension. Example: The expert consensus on the clinical diagnosis and treatment of primary aldosteronism.
2) Diabetes	Topic is related to diabetes. Example: Progress in clinical diagnosis and treatment of type 2 diabetes.
3) Gonadal disease	Topic is related to gonadal disease. Example: Delayed male hypogonadism: guidelines and controversies.
4) Neuroendocrine tumors	Topic is related to neuroendocrine tumors. Example: Long term metastatic malignant insulinoma of the liver: a case report and literature review.
5) Obesity	Topic is related to obesity. Example: Consensus on Multidisciplinary Diagnosis and Treatment of Obesity Based on Clinical Practice (2021 Edition).
6) Osteoporosis and metabolic bone disease	Topic is related to osteoporosis and metabolic disease. Example: Research progress on screening methods for osteoporosis.
7) Pituitary disease	Topic is related to pituitary disease. Example: Progress in genetic research on pituitary growth hormone adenoma.
8) Thyroid disease	Topic is related to thyroid disease. Example: Consensus among Chinese experts in the diagnosis and treatment of thyroid diseases among the elderly (2021).
Article order	
1) The first article	Article is posted in the first order of the day.
2) The second article	Article is posted in the second order of the day.
3) Other order numbers in sequence	Article is posted thereafter.
Article format	
1) Text only	Article contains only text.
2) Text + picture	Article contains text and pictures.
3) Text + picture + video	Article contains text, pictures, and videos.
4) Other	Other forms.
Length of the article	
1) length < 1k words	Number of words in the article text < 1,000.
2) length between 1k – 5k words	Number of words in the article text between 1,000 and 5,000.
3) length > 5k words	Number of words in the article text > 5,000.
Article category	
1) Review	Article provides a review of endocrine and metabolic diseases.
2) Basic Research article	Article reports basic research on endocrine and metabolic diseases.
3) Clinical Research article	Article reports clinical research on endocrine and metabolic diseases.
4) Case Report	Article reports cases of endocrine and metabolic diseases.
5) Mendelian Randomization Analysis	Article reports mendelian randomization analysis on endocrine and metabolic diseases.
6) Meta-analysis and systemic review	Article reports meta-analysis and systemic review on endocrine and metabolic diseases.
7) Guidelines	Article provides the diagnosis and treatment guidelines for endocrine and metabolic diseases.
8) Expert Consensus	Article provides the expert consensus for endocrine and metabolic diseases.
9) Others	Other categories of articles

were performed using R software (version 4.5.1). A two-sided P value ≤ 0.05 was considered statistically significant.

Results

Characteristics of WOAs of medical journals and article features of endocrine and metabolic diseases

We screened 255 medical journals from the CSCD list and, based on the inclusion and exclusion criteria, finally selected 16 medical journals that met the requirements of this study. The included 12 GMJs are Medical Journal of Peking Union Medical College Hospital, Journal of Jinan University (Natural Science & Medicine Edition), Journal of Southern Medical University, Journal of Sichuan University (Medical Science Edition), Journal of Xi'an Jiaotong University (Medical Sciences), Journal of Central South University (Medical Sciences), Journal of Chongqing Medical University, Journal of Sun Yat-sen University (Medical Sciences), Acta Academiae Medicinae Sinica, Chinese Journal of Practical Internal Medicine, Chinese Journal of Internal Medicine, and Chinese Medical Journal. The included 4 EMJs are Chinese Journal of Osteoporosis, Chinese Journal of Diabetes, Chinese Journal of Endocrinology and Metabolism, and Chinese Journal of Diabetes Mellitus.

Between 2021 and 2023, the 16 WOAs posted 1,033 specialized articles on endocrine and metabolic diseases. The most popular topics were diabetes (680/1033, 65.8%), osteoporosis and metabolic bone disease (168/1033, 16.2%), and thyroid disease (62/1033, 6.0%).Notably, 49 articles had a reading number over 10,000 times, while 1,007 articles had fewer than 10,000 times of reading. Among top-read articles, diabetes was the most common subject (32/49, 65.3%), followed by thyroid diseases (6/49, 12.2%) and osteoporosis and metabolic bone diseases (4/49, 8.2%). Among the 49 articles with a reading volume of over 10,000, the majority were guidelines and their interpretations (33 articles) or expert consensus documents (10 articles). These articles were all posted as the first article of the day, with 44 being in text-only format and 38 exceeding 5,000 words in length. The overall number of liking for the 1,033 articles was relatively low.

Table 2 presents the reading and liking metrics of public education articles according to various article characteristics for both General Medical Journal (GMJ, $N=197$) and Endocrinology and Metabolism Journal (EMJ, $N=836$).Regarding article format, significant differences were observed in GMJ for both reading (χ^2=20.63,

Table 2 Reading and liking number of different articles

	General Medical Journal ($N=197$)			Endocrinology and Metabolism Journal ($N=836$)		
	N(%)	Readings(M, IQR)	Likes(M, IQR)	N(%)	Readings(M, IQR)	Likes(M, IQR)
Article Format						
Text Only	136(69.04)	715(93.75 to 1961)	2(0 to 8)	711(85.05)	1701(1079.5 to 2892)	3(1 to 6)
Text + Pictures	55(27.92)	1372(989 to 3840)	6(2 to 10)	100(11.96)	1475.5(995 to 2468)	2(0.75 to 5)
Text + Pictures + Videos	6(3.05)	6209(3237.5 to 11414.75)	12.5(7.5 to 20.5)	25(2.99)	1709(1240 to 2154)	3(1 to 4)
Statistic		20.63	18.12		2.03	7.29
P		$P<0.01$	$P<0.01$		0.36	0.03
Length of the article						
Length < 1k words	20(10.15)	131.5(83.5 to 228)	0.5(0 to 1.75)	54(6.46)	993(675.5 to 1587.75)	2(0 to 4)
Length 1k – 5k words	146(74.11)	990(107.5 to 1894.5)	3(0 to 8)	739(88.4)	1644(1126.5 to 2555.5)	3(1 to 5)
Length > 5k words	31(15.74)	6495(2440 to 12250)	10(6.25 to 16.75)	43(5.14)	16,000(8690 to 25797.5)	17(11 to 32.5)
Statistic		55.3	36.18		120.74	104.89
P		$P<0.01$	$P<0.01$		$P<0.01$	$P<0.01$
Article category						
Review	65(32.99)	1434(523.5 to 2575.75)	7(2 to 9.75)	392(46.89)	1818.5(1187 to 2827.75)	4(2 to 6.25)
Basic Research	14(7.11)	105.5(64 to 219)	0(0 to 1)	54(6.46)	921.5(587 to 1316.75)	1(0 to 2)
Clinical Research	44(22.34)	94.5(58.25 to 919.5)	0(0 to 2)	191(22.85)	1398(913 to 2081.5)	1(0 to 3)
Case Report	17(8.63)	819(67.25 to 1352.25)	1(0 to 6.75)	105(12.56)	1562(1151 to 2141)	2(1 to 4)
Mendelian Randomization Analysis	1(0.51)	129(129 to 129)	0(0 to 0)	2(0.24)	2288.5(1605.25 to 2971.75)	1(0.5 to 1.5)
Meta-analysis and System Review	3(1.52)	1199(1079.5 to 1753.5)	5(4 to 6.5)	5(0.6)	1796(1497 to 2881)	1(0 to 5)
Guideline and Interpretation	22(11.17)	7066(3746 to 15393)	12(6.5 to 22.5)	33(3.95)	19,000(13000 to 36000)	19(12 to 35)
Expert Consensus	13(6.6)	6646(1307 to 9252)	9(3 to 16)	49(5.86)	4990(3440 to 7311)	9(7 to 15)
Others	18(9.14)	1553(928 to 2926)	5(1 to 10)	5(0.6)	2159(997 to 2493)	4(3 to 5)
Statistic		99.3	78.52		225.09	255.4
P		$P<0.01$	$P<0.01$		$P<0.01$	$P<0.01$
Article content						
Adrenal disease and Hypertension	13(6.6)	2184(652 to 3645)	7(0 to 9)	10(1.2)	1871(1200.25 to 12598.75)	3.5(2.25 to 10.25)
Diabetes	74(37.56)	1067.5(112 to 2151.75)	3(0 to 9)	589(70.45)	1766(1188 to 2937)	3(1 to 6)
Gonads	13(6.6)	1036(959 to 1378)	3(1 to 6)	12(1.44)	1874(1137.25 to 2289.5)	2(0.75 to 3.25)
Neuroendocrine Tumors	8(4.06)	755(432.75 to 4189.75)	3(0.75 to 6)	2(0.24)	1226.5(1188.75 to 1264.25)	5(3 to 7)
Obesity	19(9.64)	1275(973 to 4332.5)	6(2 to 13)	27(3.23)	1822(1262 to 2800.5)	2(1 to 4)
Osteoporosis and Metabolic Bone Disease	12(6.09)	2300(873.75 to 7670.5)	8(2 to 15.5)	156(18.66)	1286(800.75 to 2298)	3(2 to 5)
Pituitary	28(14.21)	111.5(35 to 1198.25)	0(0 to 5)	8(0.96)	1703(1167.75 to 1973.5)	1.5(0 to 2.25)
Thyroid	30(15.23)	1894.5(844 to 3476.75)	8(2 to 12.5)	32(3.83)	2443(1325.75 to 4689.75)	3.5(1 to 10.25)
Statistic		21.49	20.21		30.16	10.08
P		$P<0.01$	$P<0.01$		$P<0.01$	0.18

$P < 0.01$) and liking counts (χ^2=18.12, $P < 0.01$), with the highest median readings (6209) and likes (12.5) found in articles containing text, pictures, and videos. In EMJ, format was significantly associated with liking counts (χ^2=7.29, $P = 0.03$) but not with reading counts (χ^2=2.03, $P = 0.36$).Article length showed a significant effect in both journals (all $P < 0.01$). Articles longer than 5000 words obtained the highest median reading (GMJ: 6495; EMJ: 16000) and liking counts (GMJ: 10; EMJ: 17). Significant differences across article categories were detected for reading and liking in both GMJ (reading: χ^2=99.3; liking: χ^2=78.52; both $P < 0.01$) and EMJ (reading: χ^2=225.09; liking: χ^2=255.4; both $P < 0.01$). Guidelines/interpretations and expert consensus articles consistently achieved higher engagement metrics in both journals.For article content, reading and liking counts differed significantly across disease topics in GMJ (both $P < 0.01$). In EMJ, a significant difference was found for reading counts (χ^2=30.16, $P < 0.01$) but not for liking counts (χ^2=10.08, $P = 0.18$).

Analysis of factors affecting reading and liking volume

We conducted further analysis to examine the factors influencing the number of reading and liking and compared these factors between GMJs and EMJs. As shown in Table 3, multivariable regression analyses revealed that in GMJ, article length > 5000 words ($\beta = 1.246$, $P = 0.013$), guideline/interpretation category ($\beta = 2.825$, $P < 0.001$), and expert consensus category ($\beta = 2.715$, $P < 0.001$) were positively associated with higher reading counts. For liking counts, text+picture+video format ($\beta = 1.197$, $P < 0.001$), guideline/interpretation ($\beta = 1.936$, $P < 0.001$), and expert consensus ($\beta = 1.769$, $P < 0.001$) were significant positive predictors.In EMJ, article length > 5000 words ($\beta = 1.115$, $P < 0.001$), guideline/interpretation ($\beta = 1.710$, $P < 0.001$), and expert consensus ($\beta = 0.922$, $P < 0.001$) were associated with increased reading counts. Guideline/interpretation ($\beta = 1.888$, $P < 0.001$) and expert consensus ($\beta = 1.356$, $P < 0.001$) also positively predicted liking counts. Additionally, later article order was negatively associated with both reading (β=-0.617, $P = 0.002$) and liking counts (β=-0.900, $P = 0.006$) in EMJ.

Discussion

Endocrine and metabolic diseases pose a significant threat to human health, underscoring the importance of health education to improve public health literacy. In this retrospective study, we evaluated health education efforts by academic journals regarding endocrine and metabolic diseases through WOAs. While WOAs have become an important channel for disseminating this content, our findings revealed varying levels of engagement. Articles focusing on common endocrine diseases such as diabetes, osteoporosis, and thyroid diseases generated considerable reader interest. Notably, clinical guidelines and their interpretations achieved the highest engagement levels. Articles that incorporated images and videos garnered higher readership and greater interaction. However, the relatively low reading volumes and likes for academic articles, despite WeChat having over a billion users, suggest that there may be limitations in terms of target audience identification and the efficiency of content promotion. Further investigation into these aspects is warranted to fully understand the potential and challenges of using WOAs for health education.

New media has become an indispensable platform for public health education. Platforms such as WeChat, Facebook, and Twitter have expended into medical and patient education, providing comprehensive health information through diverse formats [7, 21, 22]. During the Covid-19 pandemic, new media platforms were crucial for disseminating health knowledge [23, 24].Concurrently, leading international medical journals, including The Lancet, The New England Journal of Medicine, Journal of the American Medical Association, British Medical Journal, etc., have actively used Twitter to promote research articles, significantly expanding their academic influence [4, 25].These journals emply various innovative strategies, such as sharing article titles and links, publishing information charts, producing podcast, and hosting online journal clubs, aiming to rapidly and widely disseminate medical research through new media. This study found that over the past three years, WOAs have served as professional platforms for publishing articles on endocrine and metabolic diseases, assuming significant responsibilities for health education among Chinese readers. Diabetes and osteoporosis have emerged as urgent public health challenges, drawing increasing attention to health education for these conditions. Traditional health education typically relies on medical staff's explanations, requiring patients and their family to visit hospitals or classrooms, or to obtain information through traditional media such as television and radio. However, with the popularization of smartphones, efficient and independent public health education models have emerged. Social media has become an important platform for health education on endocrine and metabolic diseases [26, 27]. Our data shows that the medical academic journal WOA exhibit strong engagement with endocrine and metabolic disease education, showing representative breadth in topics, audience attention and content types. For example, among the 49 articles with over 10,000 views, diabetes-themed content promineated (32/49, 65.3%), followed by thyroid diseases (6/49, 12.2%) and osteoporosis and metabolic bone diseases (4/49, 8.2%), reflecing current public health priorities. However, we noted that text remains the primary format, with insufficient integration of multimedia

Table 3 Analysis of factors affecting reading and liking volume in GMJs and EMJs

Article characteristics	General Medical Journal				Endocrinology and Metabolism Journal			
	Readings		Likes		Readings		Likes	
	Coefficient(95%CI)	P	Coefficient(95%CI)	P	Coefficient(95%CI)	P	Coefficient(95%CI)	P
Article Format								
Text Only	Reference							
Text + Picture	-0.103 (-0.453 to 0.247)	0.564	0.335 (-0.013 to 0.683)	0.061	0.059 (-0.117 to 0.235)	0.510	0.172 (-0.061 to 0.405)	0.150
Text + Picture + Video	0.489 (-0.104 to 1.082)	0.107	1.197 (0.549 to 1.845)	< 0.001	0.246 (-0.152 to 0.644)	0.227	-0.033 (-0.615 to 0.549)	0.911
Length of the article								
Length < 1k words	Reference							
Length 1k-5k words	0.224 (-0.714 to 1.162)	0.641	0.354 (-0.372 to 1.08)	0.341	0.196 (-0.081 to 0.473)	0.165	-0.346 (-0.667 to -0.025)	0.035
Length > 5k words	1.246 (0.271 to 2.221)	0.013	0.672 (-0.121 to 1.465)	0.099	1.115 (0.778 to 1.452)	< 0.001	0.348 (-0.078 to 0.774)	0.110
Article category								
Clinical Research	Reference							
Review	1.744 (0.938 to 2.550)	< 0.001	1.597 (0.973 to 2.221)	< 0.001	0.382 (0.228 to 0.536)	< 0.001	0.939 (0.697 to 1.181)	< 0.001
Meta-analysis and System Review	0.996 (-0.728 to 2.720)	0.259	1.170 (-0.198 to 2.538)	0.095	0.288 (-0.364 to 0.940)	0.387	0.077 (-1.136 to 1.290)	0.900
Case Report	0.873 (-0.224 to 1.970)	0.120	0.764 (-0.079 to 1.607)	0.077	-0.170 (-0.417 to 0.077)	0.178	0.378 (0.027 to 0.729)	0.035
Basic Research	-0.484 (-2.1 to 1.132)	0.558	-0.371 (-1.658 to 0.916)	0.573	-0.523 (-0.856 to -0.190)	0.002	-0.481 (-1.021 to 0.059)	0.081
Guideline and Interpretation	2.825 (1.965 to 3.685)	< 0.001	1.936 (1.237 to 2.635)	< 0.001	1.710 (1.469 to 1.951)	< 0.001	1.888 (1.515 to 2.261)	< 0.001
Expert Consensus	2.715 (1.844 to 3.586)	< 0.001	1.769 (1.006 to 2.532)	< 0.001	0.922 (0.714 to 1.130)	< 0.001	1.356 (1.044 to 1.668)	< 0.001
Others	1.289 (0.352 to 2.226)	0.008	1.597 (0.973 to 2.221)	< 0.001	0.201 (-0.364 to 0.766)	0.485	0.420 (-0.451 to 1.291)	0.345
Article content								
Neuroendocrine Tumors	Reference							
Diabetes	0.149 (-0.508 to 0.806)	0.657	0.749 (-0.066 to 1.564)	0.073	0.579 (-0.867 to 2.025)	0.433	-0.160 (-1.471 to 1.151)	0.811
Gonads	1.136 (0.321 to 1.951)	0.007	1.071 (0.138 to 2.004)	0.026	0.117 (-1.394 to 1.628)	0.880	-0.953 (-2.463 to 0.557)	0.216
Pituitary	0.003 (-0.893 to 0.899)	0.996	0.330 (-0.632 to 1.292)	0.503	0.423 (-1.148 to 1.994)	0.598	-0.959 (-2.730 to 0.812)	0.289
Obesity	0.947 (0.142 to 1.752)	0.022	1.364 (0.497 to 2.231)	0.002	0.499 (-0.972 to 1.970)	0.506	-0.121 (-1.493 to 1.251)	0.863
Osteoporosis and Metabolic Bone Disease	1.167 (0.379 to 1.955)	0.004	0.502 (-0.521 to 1.525)	0.337	0.074 (-1.378 to 1.526)	0.921	-0.369 (-1.687 to 0.949)	0.583
Adrenal disease and Hypertension	0.572 (-0.339 to 1.483)	0.220	0.891 (-0.059 to 1.841)	0.068	0.240 (-1.233 to 1.713)	0.749	-0.733 (-2.143 to 0.677)	0.309
Thyroid	0.499 (-0.230 to 1.228)	0.182	0.987 (0.134 to 1.840)	0.025	0.852 (-0.602 to 2.306)	0.251	0.028 (-1.307 to 1.363)	0.967
Article order	0.234 (0.089 to 0.379)	0.002	0.094 (-0.036 to 0.224)	0.155	-0.617 (-0.998 to -0.236)	0.002	-0.9 (-1.535 to -0.265)	0.006

forms such as graphics, video, and audio. In addition, the strategy for publishing tweets still needs to be improved. As vital vehicles for medical knowledge and advances, academic journals' proactive use of WOAs can rapidly dissemination health information at scale, establishing an essential channel for public health education.

Among professional articles on endocrine and metabolic diseases disseminated via WOAs, GMJs garnered lower view counts than EMJs despite similar likes counts. This discrepancy likely reflects that comprehensive medical journals, despite broad coverage, fail to attact sufficient professional readership for specilized topics. EMJs, on the other hand, focus on this field and have a clearer readership. Therefore, articles targeting endocrine and metabolic diseases can quickly reach the target readers, resulting in a higher recommended reading volume. The disparity in engagement metrics between GMJs and EMJs can be attributed to several factors. EMJs, by virtue of their specialized focus on endocrine and metabolic diseases, can attract a more targeted and engaged audience of experts and practitioners in these fields. The tailored, in-depth content of EMJs likely resonates better with their niche readership, increasing reading volumes. In contrast, GMJs, while appreciated as highlighted by the comparable liking counts, have a broader and less focused readership, which may reduce their reading engagement. This observation aligns with international trends indicating that specialized journals often achieve higher engagement within their specific areas of expertise due to a more defined readership and content relevance. However, it's essential to further investigate how factors such as posting time, promotion strategies, and reader demographics may influence these trends. Our analysis did not account for the latter two aspects, which may introduce biases and limit the generalizability of our findings.

Limitations

This study has certain limitations. First, our three-year analysis of endocrine and metabolic disease articles from medical journals' WOA may be inssuficient to capture long-term trends in the rapidly evolving new media landscape. Second, using reads and likes as engagement metrics prevented us from distinguishing between general public and medical professional audiences. Third, differences in platform influence among WOAs, such as follower numbers and posting frequency may have skewed article attention data. Forth, the inability to ascertain reader demographics and potential bias from exclusively analyzing certified WOAs, which may not represent the broader online engagement landscape. Finally, factors such as the number of followers, posting time, and article promotion were not directly included in the analysis. These factors could interact with the identified characteristics such as article format, length, and category to further impact engagement metrics. Future studies should explore these elements to provide a more comprehensive understanding of user engagement dynamics.

Conclusions

WOA is an important way for Chinese academic journals to conduct health education on endocrine and metabolic diseases. Diabetes, osteoporosis, thyroid diseases and other common chronic diseases have received extensive attention from users. The article category and the format of article have a significant impact on user engagement. The significance of this study is to provide useful references for academic journals to use new media platforms for endocrine and metabolic disease health education. By understanding the public's needs and feedback, academic journals can more accurately formulate health education strategies and content, and improve the quality and effectiveness of health education.

Abbreviations

CSCD	The chinese science citation database
EMJ	Endocrinology academic journal
GMJ	General medical journal
WOA	WeChat official account

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Authors' contributions

YL and HY designed this study, collected raw data, and completed the article writing. YD has completed data statistics and table production. NL, NZ, and YW have made revisions to the article.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Declaration of AI-assisted writing:

During the preparation of this manuscript, the authors utilized Grammarly (app.grammarly.com) and Kimi (kimi.com) solely for language editing and grammar correction to improve the readability of the text. The AI tool was not involved in study design, data collection, data analysis, or interpretation of results. The authors take full responsibility for the content, accuracy, and integrity of the manuscript. Authors confirm that all AI-generated suggestions were thoroughly reviewed, revised where necessary, and the final manuscript reflects the authors' own scientific work and conclusions.

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